Project Contribution Report

Line Follower Robot with PID Control System

Time Investment Summary

Total Project Duration: 4 days of intensive work

Lab Sessions: 12:00 PM – 8:00 PM (8 hours/day) × 3 days = 20 hours

- **Home Sessions:** 4:00 PM – 4:00 AM (12 hours/night) × 5 days = 30 hours
- **Total Team Hours:** Approximately **50 hours** combined effort

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ADHAM – Core Development & Testing

Responsibilities

- **PID Algorithm Implementation:** Designed and tuned proportional, integral, and derivative gains for line-following control
- **MATLAB GUI Development:** Built interactive control interface with real-time serial communication, parameter adjustment sliders, and error visualization
- **System Identification:** Conducted experimental data collection and performed system response analysis to characterize robot dynamics
- **Testing & Debugging:** Ran extensive field tests, identified issues, and iteratively refined PID parameters for optimal performance
- **Data Analysis:** Processed experimental data and validated control system behavior

Time Contribution

- **Lab Work:** ~20 hours (primary focus on code development, testing loops, GUI refinement)
- **Home Sessions:** ~15+ hours (late-night debugging, parameter tuning, system identification work)
- **Total: ~40+ hours** (approximately 45% of total team effort)

MOHAMED – Control System & Integration

Responsibilities

- **Arduino Firmware Development:** Programmed microcontroller to read IR sensors, compute position error, and generate PWM motor signals
- **Serial Communication Protocol:** Designed CSV logging format and implemented bidirectional serial interface with MATLAB
- **Testing Suite:** Collaborated on experimental runs, sensor calibration, and performance validation
- **System Identification:** Assisted with data collection and analysis to model robot response characteristics
- **MATLAB Integration:** Debugged serial port issues, optimized real-time plotting, and ensured synchronization with Arduino

Time Contribution

- **Lab Work:** ~20 hours (primary focus on hardware testing, sensor integration, serial debugging)
- **Home Sessions:** ~15+ hours (firmware refinement, data analysis, control tuning)
- **Total: ~40+ hours** (approximately 45% of total team effort)

ZIAD – Procurement & Assembly

Responsibilities

- **Component Sourcing:** Identified and procured all required parts (Arduino board, L298N motor driver, IR sensors, motors, power supply)
- **Hardware Assembly:** Constructed the robot chassis, mounted sensors, connected motor driver circuitry, and performed initial circuit verification
- **Testing Support:** Participated in field tests and provided feedback on hardware performance

Time Contribution

- **Lab Work:** ~10 hours (sourcing, assembly, initial setup)
- **Testing Sessions:** ~10 hours (hardware validation and troubleshooting)
- **Total: ~15 hours** (approximately 10% of total team effort)

Project Summary

This project required **distributed expertise** across firmware development, control theory, signal processing, and hardware integration. Members 1 and 2 invested significantly in the core engineering challenges—designing a functional PID controller, developing real-time visualization tools, and performing system identification to validate performance. Member 3 provided essential logistics and assembly work that enabled the team to focus on control system development without hardware delays.

The intensive 4-day sprint (80+ combined hours) resulted in a fully functional line-following robot with validated control parameters and professional-grade MATLAB interface for parameter tuning and live data visualization.

MOHAMED & ADHAM – Documentation & Presentation

Additional Responsibilities

- **Final Report:** Co-authored comprehensive project documentation including methodology, control theory, experimental results, and system identification findings
- **PowerPoint Presentation:** Created presentation slides summarizing project objectives, implementation approach, results, and conclusions for team/instructor review
- **Time Contribution:** ~10 combined hours (report writing, slide creation, data visualization)

Key Deliverables

- Functional Arduino firmware with sensor input and motor control
- MATLAB GUI with real-time PID parameter adjustment
- System identification dataset and analysis
- Tuned PID gains for stable line-following behavior
- Serial communication protocol for live monitoring
- Assembled and tested robot hardware
- Final project report with methodology and results
- PowerPoint presentation for project summary